F79SP Stochastic Processes 1 2023-24
Solutions to Exam Qns, December 2023.

Question 1. (a)

$$\begin{aligned}\mathbf{P}(S_2 \geq 0) &= \mathbf{P}(X_1 \geq 0, X_2 \geq 0) + \mathbf{P}(X_1 = -1, X_2 = 1) + \mathbf{P}(X_1 = 1, X_2 = -1) \\ &= 5/6 \cdot 5/6 + 2 \cdot 1/2 \cdot 1/6 = 31/36.\end{aligned}$$

[3 marks]

(b) Here we use the normal approximation. First, $a = \mathbf{E}X_i = 1/3$, $\sigma^2 = \mathbf{Var}X_i = 2/3 - 1/9 = 5/9$. **[1 mark]**

$$\begin{aligned}\mathbf{P}(S_{100} \geq 40) &= \mathbf{P}\left(\frac{S_{100} - 100a}{\sqrt{100\sigma^2}} \geq \frac{40 - 100a}{\sqrt{100\sigma^2}}\right) \\ &\approx 1 - \Phi((40 - 100/3)/10\sqrt{5/9}) \approx 1 - \Phi(\sqrt{0.8}) \\ &= \Phi(-\sqrt{0.8}) \approx 0.184.\end{aligned}$$

[2 marks]

(c) This is the ruin probability: equivalently, we may start from level $S_0 = 4$ **[1 mark]**
and find then the probability

$$\begin{aligned}&\mathbf{P}(\text{there exists } n > 0 \text{ such that } S_n \leq 0 \mid S_0 = 4) \\ &= \min\left(1, \left(\frac{q}{p}\right)^4\right) = \left(\frac{1/6}{1/2}\right)^4 = 1/81 \approx 0.0123.\end{aligned}$$

[2 marks]

(d) Here the probability is 1. One way to show this is to swap + and − and to apply then the formula from (c). Then p becomes $1/6$ and q becomes $1/2$, so the ratio is bigger than 1.

[2 marks]
[Total: 11 marks]

Question 2. (a) There are two closed classes: $C_1 = \{1, 5\}$ and $C_2 = \{3\}$, and one open classes $O_1 = \{2, 4\}$. Here

– C_2 is closed because it is absorbing: $p_{33} = 1$;
– C_1 is closed because $p_{11} + p_{15} = 1$, $p_{51} + p_{55} = 1$ and $p_{15}p_{51} > 0$ (the states intercommunicate);
– O_1 is an open class because $p_{24} > 0$ and $p_{42} > 0$ (so the states intercommunicate), but, say, $p_{21} > 0$, but $p_{12}^{(n)} = 0$ for all n since state 1 is in the closed class C_1.

[3 marks]

(b) State 5 belongs to the finite closed class C_1 which is aperiodic (since, say, $p_{55} > 0$).

[1 mark]

Therefore there exists a unique stationary distribution in this class, and any limiting distribution within this class coincides with the stationary one.

[1 mark]

To find the stationary distribution, we solve a system of equations:

$$\pi_5 = \pi_1 \cdot \frac{3}{4} + \pi_5 \cdot \frac{1}{3}; \quad \pi_5 + \pi_1 = 1$$

which has the unique solution $\pi_5 = 9/17$; $\pi_1 = 8/17$. So the limiting distribution for the initial state $X_0 = 5$ is $(8/17, 0, 0, 0, 9/17)$.

[3 marks]

(c) Denote by y_i the probability of absorption in $C_2 = \{3\}$ for a Markov chain that starts from $X_0 = i$.

Then $y_3 = 1$. $y_1 = y_5 = 0$,

[1 mark]

and

$$y_2 = \frac{1}{8}y_2 + \frac{1}{8} + \frac{1}{4}y_4;$$
$$y_4 = \frac{1}{6}y_2 + \frac{1}{3} + \frac{1}{6}y_4$$

[2 marks]

which leads to $y_4 = 5/11$ and $y_2 = 3/11$.

[2 marks]

(d) Let z_i be the probabilities to hit the set $\{1, 5\}$. Since the total number of states is finite, the Markov chain is always absorbed by one of closed classes (the sum of absorption probabilities is 1, for any initial state).

[2 marks]

Since the total number of closed classes is two, we get $z_4 + y_4 = 1$ and then $z_4 = 1 - y_4 = 6/11$.

[1 mark]

(e) Since the stationary distribution in the other closed class C_2 is clearly $(0, 0, 1, 0, 0)$,

[1 mark]

the limiting distribution is

$$z_4 \cdot (8/17, 0, 0, 0, 9/17) + y_4 \cdot (0, 0, 1, 0, 0) = (48/187, 0, 5/11, 0, 54/187).$$

[2 marks]

(f) This is the family

$$\gamma \cdot (8/17, 0, 0, 0, 9/17) + (1 - \gamma) \cdot (0, 0, 1, 0, 0) = (8\gamma/17, 0, 1 - \gamma, 0, 9\gamma/17)$$

where $\gamma \in [0, 1]$.

[3 marks]
[Total: 22 marks]

Question 3. (a) For that, the mean number of offspring should be not bigger than 1,

$$\mu = \mathbf{E}\xi = 2/3 \cdot 1 + (1 - x)/3 \cdot 2 \leq 1.$$

which gives $x \geq 1/2$. Also, x cannot be bigger than 1 since probabilities must be non-negative. Therefore, the answer is: $x \in [1/2, 1]$.

[2 marks]

(b) The same: $x \in [1/2, 1]$, since $\varepsilon^4 = 1$ if and only if $\varepsilon = 1$.

[2 marks]

(c) (i) There are only the following possibilities for the values of (X_1, X_2, X_3): $(1, 1, 0), (1, 2, 0), (2, 1, 0), (2, 2, 0), (2, 3, 0), (2, 4, 0)$,

[2 marks]

so the desired probability is

$$\begin{aligned} &2/3 \cdot 2/3 \cdot 1/9 + 2/3 \cdot 2/9 \cdot (1/9)^2 + 2/9 \cdot (2 \cdot 1/9 \cdot 2/3) \cdot 1/9 \\ &+2/9 \cdot ((2/3)^2 + 2 \cdot 2/9 \cdot 1/9) \cdot (1/9)^2 + 2/9 \cdot (2 \cdot 2/3 \cdot 2/9) \cdot (1/9)^3 \\ &+2/9 \cdot (2/9)^2 \cdot (1/9)^4 = 0.0563 \end{aligned}$$

[2 marks]

for the formula and

[2 marks]

for the final answer.

(ii) If the mean μ is not bigger than 1, then the extinction probability is 1. Otherwise, the extinction probability ε is the smallest (non-negative) root of equation

$$\varepsilon = \varphi(\varepsilon) \equiv \sum_k p_k \varepsilon^k.$$

[2 marks]

In our case,

$$\varepsilon = \frac{1}{9} + \frac{2}{3}\varepsilon + \frac{2}{9}\varepsilon^2,$$

or

$$\frac{2}{9}\varepsilon^2 - \frac{1}{3}\varepsilon + \frac{1}{9} = 0,$$

or

$$2\varepsilon^2 - 3\varepsilon + 1 = 0.$$

Then there are two roots, $\varepsilon_1 = 1$ and $\varepsilon_2 = 1/2$, and we choose the smallest one: $\varepsilon = 1/2$.

[3 marks]
[Total: 15 marks]

Question 4. Part (a). The proofs are discussed in Lectures.
For any $n = 0, 1, \ldots$ and $x \geq 0$, let

$$P_t(n, x) = \mathbf{P}(N(t) = n; \psi(t) > x) = \mathbf{P}(S_n \leq t; S_n + Y_{n+1} > t + x).$$

Then, for $n = 0$,

$$P_t(0, x) = \mathbf{P}(S_1 > t + x) = \mathbf{P}(Y_1 > t + x) = e^{-\alpha(t+x)} = e^{-\alpha t} \cdot e^{-\alpha x}.$$

Then, for $n > 0$,

$$\begin{aligned} P_t(n,x) &= \int_0^t f_{S_n}(z)\mathbf{P}(z + Y_{n+1} > t + x)dz \\ &= \int_0^t \frac{z^{n-1}}{(n-1)!}\alpha^n e^{-\alpha z} \cdot e^{-\alpha(t+x-z)}dz \\ &= e^{-\alpha(t+x)} \int_0^t \frac{z^{n-1}}{(n-1)!}\alpha^n dz \\ &= e^{-\alpha(t+x)} \frac{(\alpha t)^n}{n!} \\ &= \frac{(\alpha t)^n}{n!} e^{-\alpha t} \cdot e^{-\alpha x}. \end{aligned}$$

[4 marks]

Then the first marginal distribution is, for $x > 0$,

$$\begin{aligned} \mathbf{P}(\psi(t) > x) &= \sum_{n=0}^{\infty} \mathbf{P}(N(t) = n; \psi(t) > x) \\ &= e^{-\alpha x} \cdot e^{-\alpha t} \sum_{n=0}^{\infty} \frac{(\alpha t)^n}{n!} \\ &= e^{-\alpha x} \cdot e^{-\alpha t} \cdot e^{\alpha t} \\ &= e^{-\alpha x}. \end{aligned}$$

[1 mark]

The second marginal is, for $n = 0, 1, \ldots$,

$$\mathbf{P}(N(t) = n) = \mathbf{P}(N(t) = n; \psi(t) > 0) = \frac{(\alpha t)^n}{n!} e^{-\alpha t}.$$

So, we may conclude that $\psi(t)$ has an exponential distribution with parameter α and that $N(t)$ is a Poisson distribution with parameter αt.

[1 mark]

Now we show the independence. For any $n = 0, 1, \ldots$ and $x > 0$,

$$\mathbf{P}(N(t) = n; \psi(t) > x) = \frac{(\alpha t)^n}{n!} e^{-\alpha t} \cdot e^{-\alpha x} = \mathbf{P}(N(t) = n) \cdot \mathbf{P}(\psi(t) > x).$$

[1 mark]

Further, for $0 < u < t$, $(N(t) - N(u))$ and $\psi(t)$ are independent and $N(t) - N(u)$ has the same distribution with $N(t-u)$. Therefore

$$\mathbf{P}(N(t) - N(u) = n, \psi(t) > x) = \frac{(\lambda(t-u))^n}{n!} e^{-\lambda(t-u+x)}.$$

[2 marks]

Part (b).

(i) We apply the second definition of the Poisson process to conclude that this random variable has a Poisson distribution with parameter $\lambda t = 30 \cdot 4 = 120$.

[2 marks]

(ii) From the splitting theorem, this random variable has distribution $\Pi(p\lambda t) = \Pi(80)$.

[2 marks]

(iii) From lectures, the conditional distribution is Binomial, with the following parameters:

$$B\left(n, \frac{c-a}{b-a}\right)$$

where $n = 100, a = 0, c = 3$ and $b = 4$. We know that if a random variable has a Binomial distribution $B(n,p)$, its mean (expectation) is np.

[3 marks]

Thus, in our case,

$$\mathbf{E}\left(N(3\lambda) \mid N(4\lambda) = 100\right) = 100 \cdot \frac{3}{4} = 75.$$

[2 marks]
[Total: 18 marks]

Question 5. (a) There are four classes, two open and two closed: $O_1 = \{1\}$, $O_2 = \{2\}$, $C_1 = \{4\}$, $C_2 = \{3, 5\}$. Indeed, C_1 is an absorbing state since there is no chances to leave it (departure rate is 0). Then C_2 is a closed class since these two states communicate only between them, $q_{35} > 0$ and $q_{53} > 0$. Then O_2 is open since the outgoing rate is non-zero and the ingoing rate is zero. Then O_1 is open since, for example, there is a link from 2 to 4 and state 4 forms a closed class. Finally, states 1 and 2 form two different open classes since they communicate in only one direction, from 2 to 1.

[2 marks]

So the diagram contains the following links: $Q_1 \to C_1$, $O_1 \to C_2$, $O_2 \to O_1$, $O_2 \to C_2$.

[2 marks]

(b) We have to solve the systems of equations $\pi Q = \mathbf{0}$, $\sum \pi_i = 1$. There are two closed classes. The stationary distribution in the first class is clearly $(0, 0, 1, 0, 0$.

[1 mark]

The stationary distribution in the second class may be found by solving equations: $-4\pi_3 + 5\pi_5 = 0, \pi_3 + \pi_5 = 1$ which gives $(0, 0, 5/9, 0, 4/9)$.

[2 marks]

Therefore, the general form of a stationary dsitribution is $(0, 0, 5\gamma_2/9, \gamma_1, 4\gamma_2/9)$ where the γ's are non-negative and sum up to 1.

[2 marks]

(c) This is

$$\begin{pmatrix} 0 & 0 & 0 & 1/4 & 3/4 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

[2 marks]

(d) No. For the stationary distributions to be the same, the outgoing transition rates must be the same within each closed class. But, for the closed class C_2, $q_{33} = -4 \neq q_{55} = -5$.

[3 marks]

[Total: 14 marks]